

Maths

Unit 1

Beta & gamma function.

Gamma

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx$$

$$\Gamma n = (n-1)! \quad \Gamma \frac{1}{2} = \sqrt{\pi}$$

$$\Gamma n = (n-1)\Gamma(n-1)$$

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{\left[\frac{m+1}{2}\right] \cdot \left[\frac{n+1}{2}\right]}{\sqrt{\frac{m+n+2}{2}}}$$

$$\frac{\Gamma n}{a^n} = \int_0^{\infty} x^{n-1} e^{-ax} dx$$

* Relation

$$\beta(m, n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma(m+n)}$$

Beta

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\beta(m, n) = \beta(n, m)$$

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

or

$$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

*

$$\Gamma p \cdot \Gamma(1-p) = \frac{\pi}{\sin(p\pi)}$$

Area

$$A = \int_a^b 2\pi y dl$$

* General

$$A = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \quad (x \text{ Axis})$$

Parametric

$$A = \int_a^b 2\pi y(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Polar

$$A = \int_{\theta_1}^{\theta_2} 2\pi r \sin \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \quad (x \text{ Axis})$$

Volume

$$V = \int_a^b \pi y^2 dx \quad (x \text{ Axis}) \quad (y=0)$$

*

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + c$$

Unit 2

Hyperbolic function

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Convergence

Sequence

$$a_1, a_2, a_3, a_4, \dots$$

$$\lim_{n \rightarrow \infty} S_n = L$$

Series

$$S_n = a_1 + a_2 + a_3 + \dots$$

Divergence

$$\lim_{n \rightarrow \infty} S_n = \pm \infty$$

Test for Convergence

& Divergence

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^{n-1} = \frac{1}{e}$$

②

Auxiliary / Harmonic / P Series

$$\text{Series is } \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \dots$$

(i) $p > 1 \rightarrow \text{Convergence}$

(ii) $p \leq 1 \rightarrow \text{Divergence}$

Imp

if series $\rightarrow$ Convergence $\Rightarrow \lim_{n \rightarrow \infty} U_n = 0$
(Reverse is not true)

if $\lim_{n \rightarrow \infty} U_n \neq 0 \Rightarrow$ Series is Divergence
(finite)

④ Cauchy's n^{th} test

$$\lim_{n \rightarrow \infty} (U_n)^{1/n} = l \text{ (finite)}$$

$l > 1 \Rightarrow \text{Divergence}$

$l < 1 \Rightarrow \text{Convergence}$

$l = 1 \Rightarrow \text{fail}$

⑥

Raabe's test

$$\lim_{n \rightarrow \infty} n \left(\frac{U_n}{U_{n+1}} - 1 \right) = l \text{ (finite)}$$

$l > 1 \Rightarrow \text{Convergence}$

$l < 1 \Rightarrow \text{Divergence}$

$l = 1 \Rightarrow \text{fail}$

①

GP test

If series is

$$1 + x + x^2 + x^3 + \dots$$

(i) $|x| < 1 \rightarrow C$

(ii) $|x| \geq 1 \rightarrow D$

otherwise

oscillation.

③

Comparison test

$U_n = \text{Given series}$

$$\lim_{n \rightarrow \infty} \left(\frac{U_n}{V_n} \right) = l \text{ (let } V_n)$$

if V_n is Convergence
then U_n is C

if V_n is D then U_n is
also D

⑤

Ratio test

$$\lim_{n \rightarrow \infty} \left(\frac{U_n}{U_{n+1}} \right) = l$$

$l > 1 \Rightarrow \text{Convergence}$

$l < 1 \Rightarrow \text{Divergence}$

$l = 1 \Rightarrow \text{fail}$

⑦ De Morgan's (Bertrands) test

$$\lim_{n \rightarrow \infty} \left[n \left(\frac{U_n}{U_{n+1}} - 1 \right) - 1 \right] \log n = L$$

$L > 1 \Rightarrow$ Convergence

$L < 1 \Rightarrow$ Divergence

$L = 1 \Rightarrow$ fail

⑧ Logarithmic test

$$\lim_{n \rightarrow \infty} n \log \left(\frac{U_n}{U_{n+1}} \right) = L$$

$L > 1 \Rightarrow$ Convergence

$L < 1 \Rightarrow$ Divergence

$L = 0 \Rightarrow$ fail

⑨ Cauchy's Integral test

$f(x)$ = decreasing funⁿ

$$\int_1^{\infty} f(x) dx = L$$

then series $\Rightarrow$ Convergence

otherwise $\Rightarrow$ Divergence

Leibnitz's test

(for Alternating series)

Series is convergence if

(i) $U_1 \geq U_2 \geq U_3 \geq U_4 \geq \dots$

(ii) $\lim_{n \rightarrow \infty} U_n = 0$

otherwise Divergence

Power Series

$$a_0 + a_1 x + a_2 x^2 + \dots$$

(i)
$$f(x+h) = f(x) + \frac{h f'(x)}{1!} + \frac{h^2 f''(x)}{2!} + \frac{h^3 f'''(x)}{3!} + \dots$$

(ii) about point a

$$f(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \dots$$

(iii) $x \rightarrow 0$

$$f(h) = f(0) + \frac{h f'(0)}{1!} + \frac{h^2 f''(0)}{2!} + \frac{h^3 f'''(0)}{3!} + \dots$$

test

[Maclaurin's series]

$$\lim_{n \rightarrow \infty} \left(\frac{U_{n+1}}{U_n} \right) = xL \Rightarrow |xL| < 1 \Rightarrow \text{Convergence}$$

$$|xL| > 1 \Rightarrow \text{Divergence}$$

Unit 3

Even function

$$f(x) = f(-x)$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

Symmetric about y-axis

Odd function

$$f(x) = -f(-x)$$

$$\int_{-a}^a f(x) dx = 0$$

Symmetric about origin

* Fourier Series

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

here

$$a_0 = \frac{1}{2\pi} \int_c^{c+2\pi} f(x) dx$$

Euler's formulae

$$a_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \cos nx dx \quad \left| \quad b_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \sin nx dx \right.$$

* if function is even then $\Rightarrow b_n = 0$

odd then $\Rightarrow a_0 \& a_n = 0$

* Half Range Series:-

if we want only sine or only cosine series

$f^n \Rightarrow$ even $\Rightarrow b_n = 0$

Cosine series

$f^n \Rightarrow$ odd $\Rightarrow a_n = a_0 = 0$

Sine series.

Change of Interval

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \frac{\cos nx}{1} + b_n \frac{\sin nx}{1} \right]$$

$$a_0 = \frac{1}{2l} \int_{-l}^l f(x) dx$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \frac{\cos nx}{1} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \frac{\sin nx}{1} dx$$

• Ponseral theorem

$f(x) \Rightarrow$ Periodic function
(Period = $2l$)

then

$$\frac{1}{2l} \int_c^{c+2l} [f(x)]^2 dx = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Unit 4 Multivariable curves

Evaluating limits

Unit 4 (Multivariable Calculus)

Limit

Function $f(x, y)$

$$L_1 = x \rightarrow y$$

$$L_2 = y \rightarrow x$$

when $L_1 = L_2 = 0$

then check

$$L_3 (x \rightarrow mx) \text{ \& } L_4 (x \rightarrow mx^2)$$

(if $L_1 = L_2$ then
limit exist)

Continuity

if $L_1 = L_2 = f(x_0, y_0)$ then at x_0, y_0 ,
 $f(x, y)$ is continuous.

Partial Differentiation :-

$$z = f(x, y)$$

$$\frac{\partial z}{\partial x} = f_x \quad \left| \quad \frac{\partial z}{\partial y} = f_y\right.$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = f_{yx}$$

Homogenous function

$$f(x, y) = a_0 x^n y^0 + a_1 x^{n-1} y^1 + \dots + a_n x^0 y^n$$

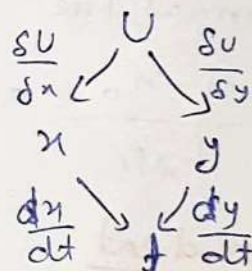
$$f(x, y) = x^n \phi\left(\frac{y}{x}\right)$$

Total Derivative

1. $U = f(x, y)$

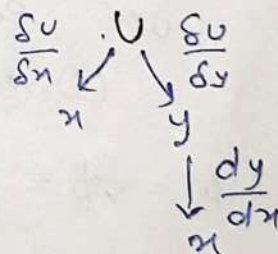
$$\frac{dU}{dt} = \frac{\partial U}{\partial x} \times \frac{dx}{dt} +$$

$$\frac{\partial U}{\partial y} \times \frac{dy}{dt}$$



(2) $U = f(x, y)$

$$\frac{dU}{dx} = \frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} \times \frac{dy}{dx}$$



Euler's theorem

$U = f(x, y)$ [homo fun]

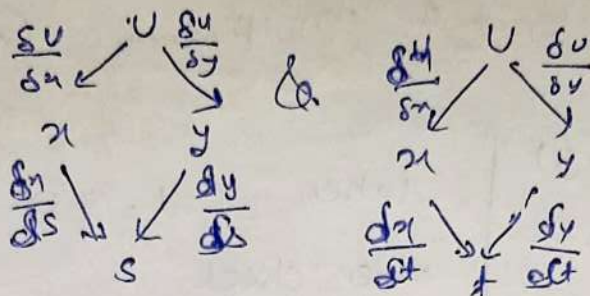
then

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f$$

$$\& \left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = n f \right)$$

$$\& \left(x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n+1)f \right)$$

3. $U = f(x, y)$



$$\frac{dU}{ds} = \frac{\partial U}{\partial x} \times \frac{dx}{ds} + \frac{\partial U}{\partial y} \times \frac{dy}{ds}$$

$$\& \quad \frac{dU}{dt} = \frac{\partial U}{\partial x} \times \frac{dx}{dt} + \frac{\partial U}{\partial y} \times \frac{dy}{dt}$$

& $f(x, y) \rightarrow$ implicit function

$$f(x, y) = 0$$

$$\frac{dy}{dx} = - \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

$\&$

* Tangent Plane & Normal

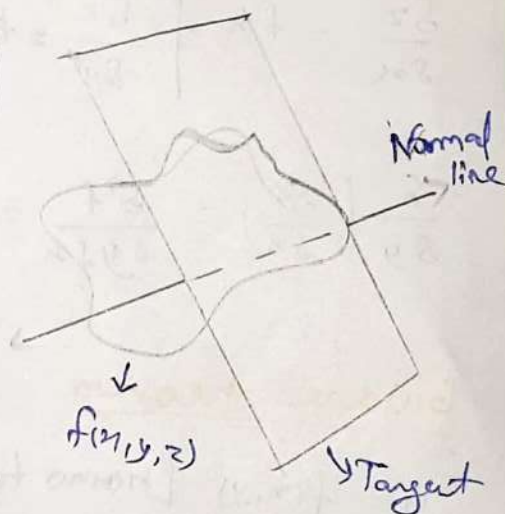
Tangent plane

$$(x-x_0)f'_x(P) + (y-y_0)f'_y(P) + (z-z_0)f'_z(P)$$

Normal line:

$$\frac{x-x_0}{f'_x(P)} = \frac{y-y_0}{f'_y(P)} = \frac{z-z_0}{f'_z(P)}$$

here $\begin{cases} f'_x = f'_1 = \frac{\partial f}{\partial x} \\ f'_y = f'_2 = \frac{\partial f}{\partial y} \end{cases}$



Gradient

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

(Direction)

Direction Derivative

for $\vec{a}$

$$DD = \nabla \phi \cdot \hat{a} = \nabla \phi \cdot \frac{\vec{a}}{|\vec{a}|}$$

$$(\text{grad}(\phi) = \nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k})$$

Divergence

(Dot product)

$$\vec{F} = f_1 \hat{i} + f_2 \hat{j} + f_3 \hat{k}$$

$$\text{div } \vec{F} = \vec{\nabla} \cdot \vec{F}$$

$$\text{div } \vec{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$\text{if } \vec{\nabla} \cdot \vec{F} = 0$$

(Solenoidal)

Curl

(Cross Product)

$$\vec{F} = f_1 \hat{i} + f_2 \hat{j} + f_3 \hat{k}$$

$$\text{Curl } \vec{F} = \vec{\nabla} \times \vec{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$\text{if } \vec{\nabla} \times \vec{F} = 0$$

(Irrotational)

• Minima & Maxima

If

$$f'(x) = f''(x) = f'''(x) = \dots = f^{(n-1)}(x) = 0$$

$$\& f^{(n)}(x) \neq 0$$

Single variable

then

n	even	$f^{(n)}(x) > 0$	local max
n	even	$f^{(n)}(x) < 0$	local min
n	odd	$f^{(n)}(x) > 0$	POI & increasing
n	odd	$f^{(n)}(x) < 0$	POI & decreasing

POI = Saddle point
point of
Inflation.

Double variable

$$f(x, y) = f$$

$$\left. \frac{\partial f}{\partial x} \right|_{p(a,b)} = 0$$

$$\left. \frac{\partial f}{\partial y} \right|_{p(a,b)} = 0$$

$$A = \begin{bmatrix} r & s \\ s & t \end{bmatrix}$$

$$(i) A_1 = |r|$$

$$(ii) A_2 = \begin{vmatrix} r & s \\ s & t \end{vmatrix}$$

$$\frac{\partial^2 f}{\partial x^2} = r \& \frac{\partial^2 f}{\partial y^2} = t \& \frac{\partial^2 f}{\partial x \partial y} = s$$

Case 1

$$\left. \begin{array}{l} A_1 \rightarrow (+) \\ A_2 \rightarrow (+) \end{array} \right\} \begin{array}{l} (+) \text{ definite} \\ \text{at } (a, b) \text{ (Minima)} \end{array}$$

Case 2

$$\left. \begin{array}{l} A_1 \rightarrow (-) \\ A_2 \rightarrow (+) \end{array} \right\} \begin{array}{l} (-) \text{ definite} \\ \text{at } (a, b) \\ \text{(Maxima)} \end{array}$$

Case 3

$$\begin{array}{l} A_1 \rightarrow 0 \\ A_2 \rightarrow 0 \end{array} \quad \begin{array}{l} \text{other than} \\ \text{(i) (ii) case} \end{array} \Rightarrow \text{Saddle Point at } (a, b)$$

Trick

$$A_1 = +$$

$$A_2 = +$$

$$A_3 = +$$

$$A_1 = -$$

$$A_2 = +$$

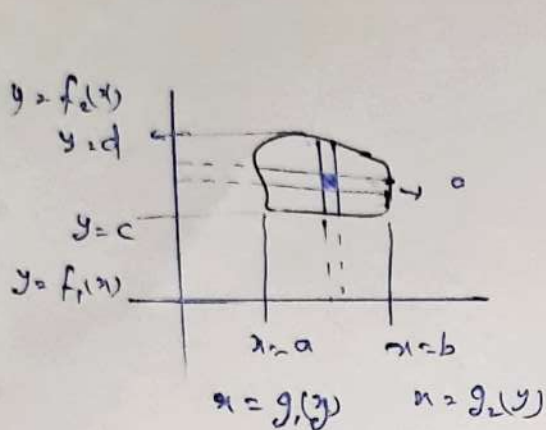
$$A_3 = -$$

Positive
(definite)
Minima

↓
negative
(maxima)

Unit 5 Multivariable Calculus

Change of Integration.



$$\int_{x=a}^b \int_{y=f_1(x)}^{f_2(x)} f(x,y) \, dy \, dx = \int_{y=c}^d \int_{x=g_1(y)}^{g_2(y)} f(x,y) \, dx \, dy$$

Area & Volume

$$A = \iint_R dx \, dy \quad \text{or} \quad \iint_R r \, dr \, d\theta$$

$$V = \iint_R f(x,y) \, dx \, dy \quad \text{or} \quad \iint_R f(r,\theta) \, r \, dr \, d\theta$$

Triple Integration

$$\iiint_S f(x,y,z) \, dx \, dy \, dz = \int_{x_1}^{x_2} \int_{y_1}^{y_2} \int_{z_1}^{z_2} f(x,y,z) \, dx \, dy \, dz$$

volume by Triple Integration.

$$\text{Vol} = \iiint_S dx \, dy \, dz$$

Line Integral

$$\text{line Integral} = \int_C \vec{f} \cdot d\vec{r}$$

$$(d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k})$$

* Surface Integral

$$\iint_S \vec{f} \cdot d\vec{S} = \iint_S \vec{f} \cdot \hat{n} ds$$

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|}$$

(here $\phi \Rightarrow$ surface)

$$ds = \frac{dx dy}{|\hat{n} \cdot \hat{k}|} = \frac{dy dz}{|\hat{n} \cdot \hat{j}|} = \frac{dx dz}{|\hat{n} \cdot \hat{i}|}$$

* Gauss Divergence Theorem

$$\iint_S \vec{f} \cdot \hat{n} ds = \iiint_V \text{div } \vec{f} dv$$

($dv = dx dy dz$)

* Centre of Gravity & COM

Density $\rho = f(x, y)$ & COM

total mass

$$m = \iint_A \rho dx dy$$

$$\bar{x} = \frac{\iint_A x \rho dx dy}{\iint_A \rho dx dy}$$

$$\bar{y} = \frac{\iint_A y \rho dx dy}{\iint_A \rho dx dy}$$

* Stokes's theorem

$$\begin{aligned} \oint_C \vec{f} \cdot d\vec{r} &= \iint_C \text{curl } \vec{f} \cdot \hat{n} ds \\ &= \iint_C (\vec{\nabla} \times \vec{f}) \cdot \hat{n} ds \end{aligned}$$

* Green's theorem

$$\begin{aligned} \oint_C \vec{f} \cdot d\vec{r} &= \oint_C M \cdot dx + N \cdot dy \\ &= \iint_S \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dx dy \end{aligned}$$